CashPilot

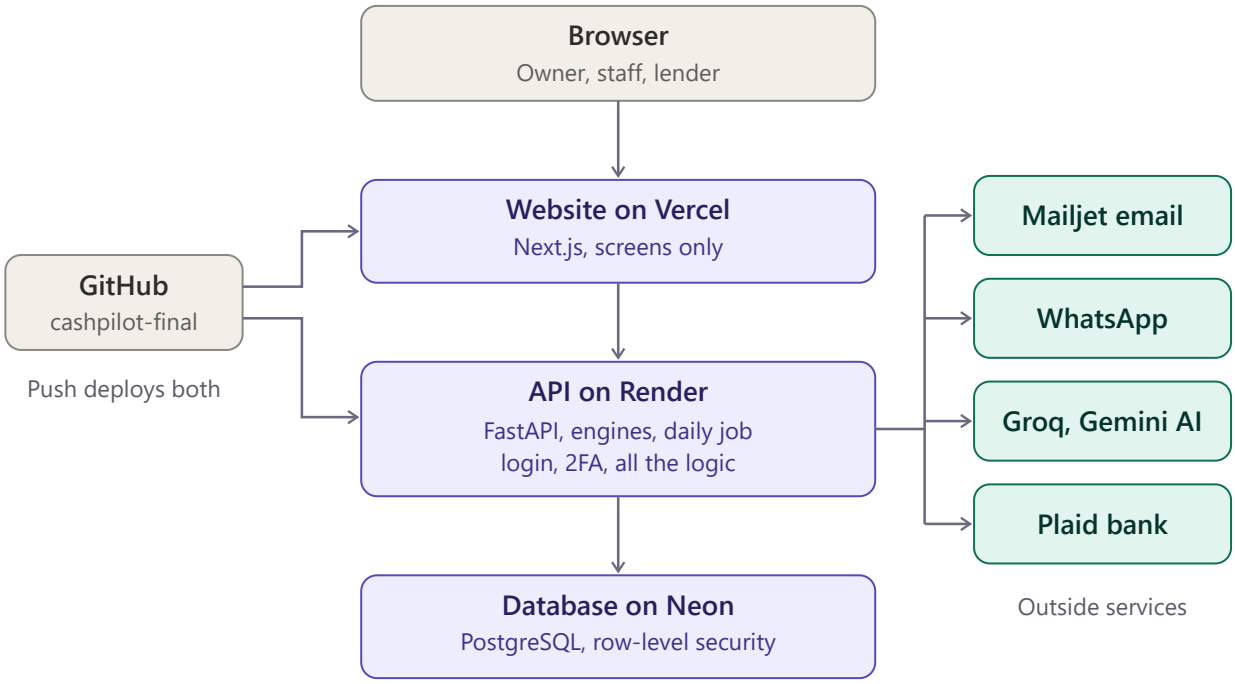
Project overview: ki ache, kothay ache, kon ta kar sathe connected

1. CashPilot ki

Choto business-er jonno AI financial assistant. Business-er records (sales, kharoch, baki, bill) theke cash shortage-er agey warning dey, ke taka dey nai ta list kore, ar credit score ar loan-er hishab dey.

> Shob number **deterministic hishab** theke ashe, AI theke na. AI shudhu bhasha-e bujhay, ar tar bola kono number engine-er sathe na mile reject hoy. Kono payment, message ba reminder owner approve na korle jay na.

2. Connection diagram



3. Kon ongsho kothay ache

Jinish	Kothay	Ki kore
Website	Vercel, `frontend/`	Next.js 16. Shudhu screen: login, dashboard, charts. Nijer kono hishab kore na, shob data API theke ane
API	Render, `backend/`	FastAPI. Login, 2FA, forecast, credit score, loan, notification, reminder pathano, shob logic ekhane

Database	Neon (PostgreSQL)	Shob data. Row-level security: ek business onno business-er data dekhte pare na
Code	GitHub, branch `cashpilot-final`	Push korle Vercel ar Render deploy hoy (Render-e Manual Deploy lage)
Email	Mailjet	Reminder email. Render free-te SMTP block, tai HTTPS. Gmail sender hole Spam-e jay
WhatsApp	Twilio sandbox ba Meta Cloud API	Owner "Send" chapole reminder pathay
AI	Groq, Gemini	Shudhu bhasha: summary, agent-er uttor. Customer/supplier name code diye dhaka thake
Bank	Plaid sandbox	Bank transaction import (Pro). Real bank na, shudhu sample
Payment	Stripe	Code ache, kintu bondho

4. Backend-er bhitor (`backend/app/`)

Folder / file	Kaj
`engines/`	Forecast, advisor, credit score, loan, collection (ke taka debe na), what-if. Shob maths ekhane
`api/`	Proti page-er endpoint: ledger, customers, notifications, finance, reports...
`alert_rules.py` , `notify.py` , `jobs/daily.py`	16 ta alert rule, notification toiri, roj-er kaj (urgent email)
`messaging.py`	Email ar WhatsApp pathano. Owner approve na korle kichu jay na
`db.py` , `models.py` , `migrations/`	Database, table, RLS
`ai/`	Groq/Gemini call, AI-r bola number engine-er sathe milano
`seed/seed.py`	Demo account (Evergreen Traders, ek bochhorer data)

5. Frontend-er bhitor (`frontend/`)

- `app/(app)/` : login-er por-er page gulo: dashboard, advisor, notifications, transactions, cashflow, ledger, customers, payables, products, credit-score, loan, reports, agent, settings.
- `app/login` , `app/signup` , `app/onboarding` , `app/page.tsx` (landing), `app/lender` (lender-er shared view).
- `components/` : AppShell (sidebar), shared UI, chart, modal, reminder dialog ityadi.
- `app/globals.css` : ForgeFlow-style dark theme.

6. Ekta kaj-er poth: reminder pathano

Browser, Vercel page, Render API (`/ledger/.../send`), Neon-e reminder save, Mailjet ba WhatsApp-e pathano, result Browser-e ferot. Provider na thakle "manual" hishebe shudhu record hoy, kono fake "sent" dekhay na.

7. Kon ongsho ekhon kon obosthay

Obostha	Ki
Chalu	Shob page, login, 2FA, notification, email (spam folder-e), Plaid sandbox, credit score ar loan, customer edit
Aadha	WhatsApp (Twilio/Meta-er shorto baki), demo account live-e (seed code deploy hoyeche, kintu account ekhono toiri hoy ni)

Bondho / baki	Stripe payment, real lender API, real bank (shudhu sandbox)

8. Live address

Ki	Address
Website	https://chinese-hackathon.vercel.app (ba notun alias https://cashpilot-ai-demo.vercel.app)
API	https://cashpilot-api-26h3.onrender.com
Code	github.com/GaziArnob/chinese-hackathon, branch cashpilot-final
Demo login	demo@example.com / Demo@12345 (local-e chalu; live-e demo seed hole)

> **Secret:** database password, API token, key shudhu Render-er Environment-e thake, code ba GitHub-e na. Kono jaygay screenshot-e ei value dekha gele sheta rotate (notun kore) nite hoy.